Rectifier

Rectifier is a device which converts the sinusoidal AC voltage into either positive or negative pulsating DC.

Rectifier output contains unwanted AC components called as ripples. Diodes rectify the signal.

Filter

Filter is a device which passes DC components and blocks AC components of the rectifier output.

Resistor, capacitor and inductor (passive components) are used for this purpose.

Regulator

Regulator or stabilizer is a device that is used to maintain a constant DC output voltage.

Zener diode is used as regulator.

4.3. RECTIFIER

Figure 4.2 shows the schematic diagram of a Rectifier



Figure 4.2 Rectifier

Semiconductor diode (mostly silicon diode) is used as a rectifier element. It converts alternating current into direct current. Hence let we see the working of semiconductor diode.

4.3.1 Semiconductor Diode

A PN junction is known as a semi-conductor or crystal diode.

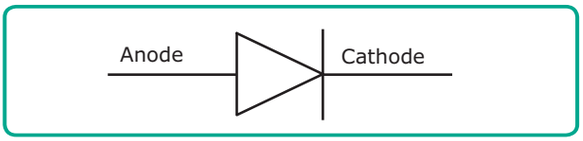


Figure 4.3 Symbol of diode

The character of a crystal diode is to conduct current in one direction only, permits it to be used as a rectifier. A crystal diode is usually represented by the schematic symbol shown in figure 4.3. The arrow in the symbol indicates the direction of conventional current flow.

The crystal diode has two terminals. When it is connected in a circuit, one thing to decide is whether the diode is forward or reverse biased. There is an easy rule to ascertain it. If the external circuit is trying to push the conventional current in the direction of arrow, the diode is forward biased. On the other hand, if the conventional current is trying to flow opposite to arrow head the diode is reverse biased. Putting in simple words:

1. If arrow head (i.e., Anode) of diode symbol is positive with respect to bar (i.e., Cathode) of the symbol the diode is forward biased.
2. If the arrow head (Anode) of diode symbol is negative with respect to bar(Cathode), the diode is reverse biased.

4.3.2 Identifying the Terminals of Diodes

The Anode or Cathode of the diode is clearly marked on the body or envelop. The cathode of the diode which is in the glass envelop is marked by a bar on the body. This is shown in figure 4.4.

ELEC_EQUIP_Chapter04.indd 85 11/10/2020 11:38:45 AM